

SARNY, Ukraine

WMO: 330880

Lat: 51.3128N

Lon: 26.6139E

Elev: 512

StdP: 14.43

Time Zone: 2.00 (EUE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-1.9	4.8	-6.4	4.0	-0.7	-0.3	5.5	6.3	25.4	34.1	22.0	33.2	2.3	340	0.486

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.1	87.6	68.8	84.4	67.2	81.4	65.9	71.0	82.9	69.3	80.1	67.6	77.4	5.6	50

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
67.2	102.1	75.1	65.7	96.5	73.6	64.1	91.4	72.1	35.1	83.0	33.7	80.2	32.3	77.2	77.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4)	18.9	15.3	12.3	DB	-8.4	91.9	7.0	3.0	-13.4	94.1	-17.5	95.9	-21.4	97.6	-26.5	99.8
(5)				WB	-8.7	73.5	6.9	1.8	-13.7	74.8	-17.7	75.9	-21.6	76.9	-26.6	78.2

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	47.2	25.8	28.2	36.0	48.5	58.1	64.2	67.8	66.1	56.9	46.7	37.7	29.0	(6)
(7)		DBStd	16.89	10.52	10.25	8.45	7.70	7.47	6.17	5.81	5.89	6.92	7.55	8.34	9.73	(7)
(8)		HDD50	3129	751	610	436	116	17	0	0	17	156	376	650	(8)	
(9)		HDD65	6819	1216	1030	898	495	235	90	38	57	256	569	820	1115	(9)
(10)		CDD50	2100	0	0	3	72	269	426	550	499	224	52	5	0	(10)
(11)		CDD65	318	0	0	0	1	23	66	123	91	14	0	0	0	(11)
(12)		CDH74	3202	0	0	0	25	284	639	1171	937	144	2	0	0	(12)
(13)		CDH80	932	0	0	0	5	59	157	377	306	28	0	0	0	(13)
(14)	Wind	WSAvg	4.3	4.9	4.9	5.1	4.4	4.2	4.0	3.6	3.4	3.7	4.3	4.5	4.5	(14)
(15)	Precipitation	PrecAvg	24.4	1.3	1.3	1.4	1.5	2.4	2.8	4.3	2.4	2.2	1.6	1.8	1.6	(15)
(16)		PrecMax	30.9	3.0	2.2	3.2	3.8	5.7	5.9	10.3	7.0	4.6	5.1	4.0	2.8	(16)
(17)		PrecMin	17.2	0.3	0.7	0.2	0.2	0.8	0.5	1.1	0.1	0.3	0.2	0.2	0.6	(17)
(18)		PrecStd	3.6	0.6	0.4	0.7	0.7	1.0	1.2	2.8	1.7	1.3	1.1	0.8	0.7	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	47.0	51.3	62.5	77.5	85.6	88.2	91.5	92.0	83.2	72.1	59.7	48.5	(19)
(20)			MCWB	42.4	45.7	49.6	59.7	64.3	68.4	70.7	69.3	65.1	59.6	53.2	46.1	(20)
(21)		2%	DB	43.2	46.0	56.9	70.8	81.1	84.5	87.7	87.2	78.0	66.8	55.0	45.3	(21)
(22)			MCWB	40.2	41.9	46.6	56.2	64.0	66.8	69.9	68.6	62.8	57.3	51.2	43.1	(22)
(23)		5%	DB	40.4	43.0	52.4	66.4	76.9	81.2	84.3	83.1	73.7	61.8	51.6	42.8	(23)
(24)			MCWB	38.1	39.7	44.9	53.7	62.0	65.4	68.1	67.0	61.2	55.0	49.1	41.0	(24)
(25)		10%	DB	37.6	39.9	48.2	62.4	72.7	77.6	80.9	79.1	69.4	58.2	48.8	40.2	(25)
(26)			MCWB	35.9	37.4	42.5	51.7	59.3	63.9	66.6	65.3	59.1	53.1	46.6	38.4	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	43.4	46.2	51.6	60.9	67.8	71.0	73.7	73.0	68.4	61.9	54.4	46.9	(27)
(28)			MDCB	45.9	50.1	59.5	73.5	80.1	82.8	85.3	85.2	80.2	68.8	57.9	48.5	(28)
(29)		2%	WB	40.7	42.8	48.3	57.6	64.9	69.0	71.5	70.5	64.7	58.8	51.8	43.6	(29)
(30)			MDCB	42.8	45.3	54.3	68.0	77.2	79.9	83.7	82.0	73.7	64.6	54.4	45.1	(30)
(31)		5%	WB	38.3	40.1	45.6	55.0	63.2	67.3	69.8	68.5	62.4	56.2	49.3	41.2	(31)
(32)			MDCB	40.2	42.4	51.2	64.5	74.6	77.4	81.1	79.0	70.7	60.9	51.6	42.6	(32)
(33)		10%	WB	36.1	37.5	43.1	52.8	61.2	65.7	68.1	66.7	60.6	53.5	46.7	38.6	(33)
(34)			MDCB	37.5	39.7	47.9	60.5	70.4	74.6	77.3	76.7	68.2	57.4	48.5	40.0	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	8.4	10.3	13.6	18.3	20.0	19.1	19.1	20.4	17.6	14.6	8.9	8.0	(35)
(36)			MCDBR	8.7	12.5	20.9	25.1	26.3	25.0	24.9	26.7	24.7	21.1	14.1	10.0	(36)
(37)			MCWBR	7.4	9.3	12.6	12.4	11.4	9.9	9.2	10.7	11.4	12.2	9.8	8.2	(37)
(38)		5% WB	MCDBR	8.0	11.6	18.6	22.7	23.2	21.2	21.9	22.3	20.8	18.4	12.3	9.6	(38)
(39)			MCWBR	7.1	9.2	11.9	12.1	11.2	9.6	9.1	10.0	11.0	11.7	9.2	8.6	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.309	0.342	0.376	0.427	0.413	0.404	0.434	0.433	0.401	0.361	0.344	0.307	(40)	
(41)		taud	2.211	2.163	2.196	2.153	2.267	2.316	2.241	2.238	2.280	2.369	2.304	2.258	(41)	
(42)		Ebn at Noon	216	241	257	256	267	271	259	252	246	235	203	198	(42)	
(43)		Edh at Noon	23	31	37	43	40	39	41	39	34	26	21	19	(43)	
(44)	All-Sky Solar Radiation	RadAvg	256	510	896	1329	1684	1818	1715	1525	1038	593	245	174	(44)	
(45)		RadStd	41	84	133	132	158	135	140	112	148	103	23	41	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+1.31	N/A	N/A	+1.75	+1.04	+0.85	-272	-418	+202	+72
(47)	Regional (7 neighbors)	+1.15	N/A	N/A	+1.57	+1.10	+1.03	N/A	-355	+202	+79

Nomenclature: See separate page